

MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 14 (22-26 Nov 2026) - Period 1	Unit 4 - 4.3 Game Enhancements
Pacing (2-1-1-2): ☒ On plan ☐ Ahead ☐ Behind Weeks ahead / behind: 0 Week / lesson: Week 14 / Lesson 1 of 2		Starting point (2-2-1-3): What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it? Bug-fix homework: 75% fixed one bug; playtest logs good. Today: improve player experience with a custom cursor, custom popup and sound.		

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — trace to the curriculum / standard and the table of specifications (ToS)

Students customise the cursor (glove), replace alert() with a styled popup box, and add audio (background music, catch/game-over sounds, mute). Sprint 4.3 obj. 1-4; CSTA 3A-AP-19, 3A-AP-21.

Key knowledge / skill / understanding:

cursor: url() with fallback; classList show/hide; new Audio(), play(), volume; autoplay blocked until a click.

Success criteria — what will successful learning look / sound like?

I set a custom cursor with a fallback; my results popup shows and closes with a button; a sound plays when a worm is caught; players can mute.

Likely misconception / access or challenge need:

Any image size works as a cursor; alert() can be styled; music can autoplay on load.

2 PROOF — What will students do to show they have learned it? Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 student task + evidence of learning	ACTIVITY 2 student task + evidence of learning	ACTIVITY 3 student task + evidence of learning
Strategy: Discussion / dialogue ☐ Time: 5 min	Strategy: Modelling / worked exam ☐ Time: 25 min	Strategy: Role play ☐ Time: 6 min
What students will do: Compare alert() vs a custom popup and silent vs sound versions of the game; agree 3 UX rules (clear, not annoying, player in control).	What students will do: Code along in 3 steps: 1) glove cursor (32x32 png); 2) results popup (overlay div, message, Restart button, classList toggle); 3) catch and game-over sounds + mute button.	What students will do: Player and developer: player tests the enhancements and gives feedback (too loud, popup unclear); developer makes one change and notes it in the change log.
Evidence tool: Oral questioning ☐ Assessment: ☒ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☐ Pair ☒ Group	Evidence tool: Practical / performance task ☐ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☒ Individual ☐ Pair ☐ Group	Evidence tool: Peer-assessment checklist ☐ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☐ Individual ☒ Pair ☐ Group
Resources: Two demo builds of the game.	Resources: VS Code; QR 'Customize the cursor', 'Create a customized popup', 'Audio elements'; project sources.	Resources: Feedback checklist; headphones.
Support (low performers / SEN): Picture comparison with key words.	Support (low performers / SEN): Popup HTML/CSS and audio files ready; students write the JS lines only.	Support (low performers / SEN): Feedback phrases card.
Stretch (fast learners / gifted): Name an accessibility risk of sound-only feedback.	Stretch (fast learners / gifted): Challenge 1 audio player (play/pause/volume) or Challenge 3 image carousel.	Stretch (fast learners / gifted): Developer writes a change log entry with reason.

3 PREPARE — What do I need to do so students can succeed?	4 RESPOND — What will I do with what I see and hear?
Set students up to succeed: ☒ Hook / prior learning ☒ Explain or model ☒ Success criteria shared ☒ Key vocabulary / examples ☒ Real-life connection ☒ Clear instructions / resources ☒ Praise, recognition and reinforcement planned	If students struggle or finish early I will: ☒ Re-teach ☒ Regroup ☒ Give extra support ☒ Stretch ☒ Have students redo / improve
My setup / modelling: Time: 7 min Hook: play the game - the alert box interrupts and it is silent. Model CSS cursor with fallback; popup structure and classList.add/remove('show') with z-index; new Audio('catch.mp3').play() inside the click handler; mute toggle. Remind headphones and respectful volume.	What I will watch for / likely responses: Cursor image too large (ignored); popup hidden behind elements (z-index); autoplay error in Console; wrong audio path; overlapping sounds (reset currentTime). Likely: 'my cursor/sound doesn't work' - check size/path.
What is new or innovative in this lesson? <i>(ETEC looks for this)</i> User-centred testing through player/developer role play; inclusive design with a mute control and visual feedback alongside sound.	Close / exit check <i>(summative + time for learners' questions)</i> Time: 7 min Exit ticket: write the CSS for a custom cursor with fallback; why must audio start after a click? Learners' questions.
Progress recorded / shared: ☒ Gradebook ☐ Parents ☒ Written feedback in notebook / portfolio	Homework / platform <i>(e.g. Madrasati)</i> Madrasati: upload your game (zip) with cursor, popup and sound. Bring one animation idea for the gloves or worms.

5 REFLECT — After the lesson <i>Complete within a day. This is reflective-practice evidence and feeds your next Starting point.</i>		
What worked — and how do I know? Three enhancements in one period worked because popup and audio starters were ready; 75% finished all three. Students chose music thoughtfully for peers from different backgrounds.	What will I change next time? Headphones for every PC; testing sound in a full room was noisy. Pre-check audio paths in each project folder.	Students needing follow-up: 10/2: students absent last week need the fixed core-logic source; 2 SEN students need the popup template on paper.

COVERAGE CHECK <i>Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.</i>							
Literacy read + write · 2-1-1-6 ☒ 1 ☐ 2 ☐ 3 ☒ C ☐ H	Numeracy numbers / data · 2-1-1-6 ☐ 1 ☐ 2 ☐ 3 ☐ C ☒ H	Digital students use it · 2-1-1-4 ☐ 1 ☒ 2 ☒ 3 ☐ C ☒ H	Digital research trusted sources · 2-1-1-4 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Real-life ① application · 2-1-1-5 ☒ 1 ☐ 2 ☐ 3 ☐ C ☐ H	Real-life ② application · 2-1-1-5 ☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	Inquiry experiment / research · 2-1-1-7 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Thinking problem-solving · 2-1-1-7 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H
Create something new · 2-1-1-7 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Discussion expressing ideas · 2-1-1-8 ☒ 1 ☐ 2 ☒ 3 ☐ C ☐ H	Pair / group cooperative work · 2-1-1-8 ☒ 1 ☐ 2 ☒ 3 ☐ C ☐ H	Learner choice independent work · 3-2-1-6 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Self-assessment 2-2-1-1 ☐ 1 ☐ 2 ☐ 3 ☒ C ☐ H	Peer assessment 2-2-1-1 ☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	SEN / low specific support · 2-1-1-3 ☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H	Gifted specific stretch · 2-1-1-3 ☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback
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Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 14 (22-26 Nov 2026) - Period 2	Unit 4 - 4.4 Animations + Term 1 showcase
Pacing (2-1-1-2): ☒ On plan ☐ Ahead ☐ Behind Weeks ahead / behind: 0 Week / lesson: Week 14 / Lesson 2 of 2		Starting point (2-2-1-3): What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it? Games with cursor/popup/sound uploaded by ~80%; mute done by 60%. Final period of Term 1: CSS animations, then a short showcase.		

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — trace to the curriculum / standard and the table of specifications (ToS)

Students add hover transitions and CSS @keyframes animations (glove swing, popup scale-in) triggered from JavaScript, finalise the worm game and show it to peers. Sprint 4.4 obj. 1-4; CSTA 3A-AP-16, 3A-AP-23; ISTE 6.d.

Key knowledge / skill / understanding:

transition = 2 states; @keyframes defines stages in %; add/remove a class in JS to replay; animationend event.

Success criteria — what will successful learning look / sound like?

I add :hover transitions to buttons; I write a 3-stop @keyframes animation; my glove animates when catching a worm; I explain one design decision to a visitor.

Likely misconception / access or challenge need:

Animations need JavaScript; % in keyframes means size; replaying needs a page refresh.

2 PROOF — What will students do to show they have learned it? Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 student task + evidence of learning	ACTIVITY 2 student task + evidence of learning	ACTIVITY 3 student task + evidence of learning
Strategy: Modelling / worked exam ☐ Time: 12 min	Strategy: Experiment / lab ☐ Time: 12 min	Strategy: Stations / carousel ☐ Time: 13 min
What students will do: Code along: button hover (transform: scale + transition); glove @keyframes swing triggered by adding .catch on click and removing it on animationend; popup scale-in.	What students will do: Animation lab: tweak duration, timing function and iteration of your glove animation; record what changes. Optional: Challenge 3 rectangle-to-circle.	What students will do: Mini-showcase: half the class hosts their game, half visit and play, then swap; visitors score with the Forms peer rubric.
Evidence tool: Mini-whiteboards ☐ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☒ Individual ☐ Pair ☐ Group	Evidence tool: Portfolio / notebook work ☐ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☐ Individual ☒ Pair ☐ Group	Evidence tool: Peer-assessment checklist ☐ Assessment: ☐ Diagnostic ☐ Formative ☒ End Grouping: ☐ Individual ☐ Pair ☒ Group
Resources: VS Code; QR 'CSS animations'; project CSS source.	Resources: Lab record sheet; Challenge 3 page.	Resources: Forms peer rubric via QR on each PC; host script card.
Support (low performers / SEN): Ready CSS; students add one line of JS.	Support (low performers / SEN): Record sheet with 2 rows pre-filled.	Support (low performers / SEN): Host script card to explain the game.
Stretch (fast learners / gifted): Use cubic-bezier timing; add prefers-reduced-motion.	Stretch (fast learners / gifted): Challenge 1: three vehicles moving in a square path.	Stretch (fast learners / gifted): Host explains the code of one feature.

3 PREPARE — What do I need to do so students can succeed?				4 RESPOND — What will I do with what I see and hear?			
Set students up to succeed: ☒ Hook / prior learning ☒ Explain or model ☒ Success criteria shared ☒ Key vocabulary / examples ☒ Real-life connection ☒ Clear instructions / resources ☒ Praise, recognition and reinforcement planned				If students struggle or finish early I will: ☒ Re-teach ☒ Regroup ☒ Give extra support ☒ Stretch ☒ Have students redo / improve			
My setup / modelling: Time: 5 min Hook: our game vs the same game with bouncing buttons and a swinging glove. Model transition vs animation; write @keyframes with 0%, 50%, 100%; show the class-toggle trigger and animationend. Explain showcase routine and respectful feedback; save a backup before changes.				What I will watch for / likely responses: Animation name mismatch; no duration so nothing moves; animation plays once only; last-minute changes breaking the game (backup!); vague peer feedback. Likely: 'it worked before!'			
What is new or innovative in this lesson? <i>(ETEC looks for this)</i> Lab-style experimenting with animation variables, then an end-of-term showcase where every student is both host and reviewer.				Close / exit check <i>(summative + time for learners' questions)</i> Time: 8 min Summative reflection (Forms): what I learned in Units 2-4, my best feature, my next goal. Learners' questions and celebration.			
Progress recorded / shared: ☒ Gradebook ☒ Parents ☒ Written feedback in notebook / portfolio				Homework / platform <i>(e.g. Madrasati)</i> Madrasati: upload final zipped game + 3-line README of design decisions. Share the game with family and note one suggestion.			

5 REFLECT — After the lesson <i>Complete within a day. This is reflective-practice evidence and feeds your next Starting point.</i>		
What worked — and how do I know? Animations made the games feel finished; showcase built confidence, with peers translating for newer learners. 85% of games ran fully.	What will I change next time? Animation + showcase in one period was tight; next year give the showcase its own period. Collect backups earlier in the week.	Students needing follow-up: Students who did not finish (2-4 per class): extension to Term 2 week 1; use high achievers' games as exemplars.

COVERAGE CHECK <i>Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.</i>							
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